

MHSDS KPI Extraction and Analysis Report

Dataset used: ↓ [MHSDS Monthly — NHS Digital](#)

WEEK 4 ----- ANA 1

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PROJECT NAME: AI MENTAL HEALTH UK

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1.MHSDS KPI Extraction and Analysis:

This task focuses on analysing important NHS mental health indicators using the official MHSDS dataset. The analysis was performed to understand the performance and utilization of different NHS mental health services by extracting key KPI measures from the dataset.

2. Objective:

The main objective of this task was to extract and analyse important NHS Mental Health KPIs from the official Mental Health Services Data Set (MHSDS). The task aimed to identify important service indicators related to contacts, admissions, bed occupancy, crisis services, and waiting-time measures. The analysis also focused on understanding overall service demand and operational trends within NHS mental health systems using data visualization and KPI summaries.

3. Dataset Information:

Dataset Used:

MHSDS Monthly: Time Series for Selected Measures April 2016 to Performance February 2026

Source:

NHS Digital – Mental Health Services Data Set (MHSDS)

Dataset Description:

The MHSDS dataset contains NHS mental health service performance indicators collected across different NHS regions and service categories. The dataset includes measures related to patient contacts, inpatient admissions, ward occupancy, crisis referrals, and treatment waiting times.

The dataset provides valuable information about mental health service usage and operational performance across NHS mental health systems. Different KPI measures were extracted using keyword-based filtering techniques from the MEASURE_NAME column.

4. Tools and Technologies Used:

The analysis was performed using Python-based data analytics tools in Google Colab environment.

Tools Used:

- Python
- Google Colab
- Pandas
- Matplotlib
- CSV Dataset
- NHS Open Data

Purpose of Tools:

- Pandas was used for data cleaning, filtering, and KPI extraction.
- Matplotlib was used for generating charts and visualizations.
- Google Colab was used for performing the complete analysis in an interactive notebook environment.
- NHS open datasets were used as the primary data source for the analysis.

5. Data Preprocessing:

Several preprocessing steps were performed before analysing the dataset to ensure proper extraction and visualization of KPI measures.

Preprocessing Steps:

1. The CSV dataset was loaded into a Pandas DataFrame using `pd.read_csv()`.
2. Dataset structure and columns were checked using `head()` and column inspection methods.
3. The `MEASURE_NAME` column was used to identify KPI-related measures.
4. Separate DataFrames were created for Contacts KPI, Admissions KPI, Beds KPI, RTT KPI, and Crisis KPI.
5. Keyword-based filtering techniques were applied using `str.contains()`.
6. The `MEASURE_VALUE` column was converted into numeric format using `pd.to_numeric()`.
7. Invalid and non-numeric values were handled using `errors='coerce'`.
8. Final KPI summaries were generated after cleaning and aggregation.

These preprocessing steps helped in preparing the dataset for accurate KPI analysis and visualization.

6. KPI Extraction:

Five important NHS mental health KPIs were extracted from the dataset using keyword-based filtering on the `MEASURE_NAME` column.

The extracted KPIs were:

- Contacts KPI
- Admissions KPI
- Beds KPI
- RTT KPI
- Crisis KPI

Each KPI represents a different operational aspect of NHS mental health services.

6.1 Contacts KPI

Keywords Used:

Contact

Purpose:

The Contacts KPI was extracted to analyse overall mental health service interactions and patient contacts across NHS mental health systems.

Analysis:

The dataset showed a very high number of contacts-related records, indicating strong utilization of NHS mental health support services. High contact values suggest that a large number of patients are regularly interacting with NHS mental health services for consultations, assessments, and treatment support.

This KPI provides important insight into overall mental health service demand and accessibility.

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	STATUS	BREKDOWN	PRIMARY_LEVEL	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
3	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00L	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00L	NONE	NONE	NHS29	Contacts in RP	15330
4	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00L	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00L	NONE	NONE	NHS99	The number of children and young people, regar...	705
5	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00N	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00N	NONE	NONE	NHS01	People in contact with services at the end of ...	9045
10	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00N	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00N	NONE	NONE	NHS29	Contacts in RP	12635

6.2 Admissions KPI:

Keywords Used:

inpatient, admitted, hospital

Purpose:

The Admissions KPI was extracted to analyse inpatient mental health admissions and hospital-related service activity.

Analysis:

The extracted admissions-related measures represent patients requiring hospital-based mental health care and inpatient support. Compared to contacts and crisis measures, admissions values were lower, which is expected because inpatient treatment is generally required for more severe mental health conditions.

This KPI helps in understanding the level of inpatient mental health burden across NHS services.

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	STATUS	BREAKDOWN	PRIMARY_LEVEL	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
6	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00N	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00N	NONE	NONE	MHS07	People with an open hospital spell at the end ...	75
13	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00P	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00P	NONE	NONE	MHS07	People with an open hospital spell at the end ...	110
20	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00Q	NHS LANCASHIRE AND SOUTH CUMBRIA ICB - 00Q	NONE	NONE	MHS07	People with an open hospital spell at the end ...	85
27	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00R	NHS LANCASHIRE AND SOUTH CUMBRIA ICB - 00R	NONE	NONE	MHS07	People with an open hospital spell at the end ...	90

6.3 Beds KPI:

Keywords Used: bed, ward.

Purpose

The Beds KPI was extracted to analyse bed occupancy and ward utilization within NHS mental health facilities.

Analysis:

The analysis identified substantial bed-related activity, showing the importance of inpatient infrastructure in mental health treatment. High bed occupancy measures indicate significant utilization of mental health wards and inpatient care facilities.

This KPI is useful for understanding mental health resource allocation and inpatient service pressure.

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	STATUS	BREAKDOWN	PRIMARY_LEVEL	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
0	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00L	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00L	NONE	NONE	MHS24a	Under 16 bed days on adult wards in reporting ...	*
1	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00L	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00L	NONE	NONE	MHS24b	Age 16 bed days on adult wards in reporting pe...	*
2	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00L	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00L	NONE	NONE	MHS24c	Age 17 bed days on adult wards in reporting pe...	*
7	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00N	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00N	NONE	NONE	MHS24a	Under 16 bed days on adult wards in reporting ...	*
8	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00N	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00N	NONE	NONE	MHS24b	Age 16 bed days on adult wards in reporting pe...	*
9	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00N	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00N	NONE	NONE	MHS24c	Age 17 bed days on adult wards in reporting pe...	*

6.4 RTT KPI:

Keywords Used:

waiting, RTT

Purpose:

The RTT (Referral to Treatment Time) KPI was extracted to analyse waiting-time performance and referral delays within NHS mental health systems.

Analysis:

The RTT measures highlighted waiting-time pressures and ongoing referral management challenges in mental health services. These indicators help identify delays between referrals and treatment initiation. RTT analysis is important for evaluating healthcare efficiency and patient service accessibility.

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	STATUS	BREAKDOWN	PRIMARY_LEVEL	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
48461	01/02/2025	30/04/2025	Performance	England	England	England	NONE	NONE	ED08	Referrals with eating disorder issues waiting...	5362
48462	01/02/2025	30/04/2025	Performance	England	England	England	NONE	NONE	ED09	Referrals with eating disorder issues categori...	728
48463	01/02/2025	30/04/2025	Performance	England	England	England	NONE	NONE	ED09a	Referrals with eating disorder issues categori...	57
48464	01/02/2025	30/04/2025	Performance	England	England	England	NONE	NONE	ED09b	Referrals with eating disorder issues categori...	65
48465	01/02/2025	30/04/2025	Performance	England	England	England	NONE	NONE	ED09c	Referrals with eating disorder issues categori...	122

6.5 Crisis KPI:

Keywords Used:

referral, crisis

Purpose:

The Crisis KPI was extracted to analyse urgent mental health referrals and crisis-related service demand.

Analysis:

The crisis-related measures recorded significantly high values, indicating increased demand for urgent mental health interventions and emergency support services. High crisis referral counts may reflect growing mental health concerns and increased pressure on NHS crisis management systems. This KPI provides important insights into emergency mental health care demand across NHS regions.

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	STATUS	BREAKDOWN	PRIMARY_LEVEL	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
4	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00L	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00L	NONE	NONE	MHS69	The number of children and young people, regar...	705
11	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00N	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00N	NONE	NONE	MHS69	The number of children and young people, regar...	800
18	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00P	NHS NORTH EAST AND NORTH CUMBRIA ICB - 00P	NONE	NONE	MHS69	The number of children and young people, regar...	1270
25	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00Q	NHS LANCASHIRE AND SOUTH CUMBRIA ICB - 00Q	NONE	NONE	MHS69	The number of children and young people, regar...	715
32	01/04/2025	30/04/2025	Performance	Sub ICB - GP Practice or Residence	00R	NHS LANCASHIRE AND SOUTH CUMBRIA ICB - 00R	NONE	NONE	MHS69	The number of children and young people, regar...	600

7. KPI Summary Table

After extracting all KPI categories, the total values were calculated and summarized. The summary table clearly shows that Contacts KPI and Crisis KPI recorded the highest overall values among all extracted indicators.

	KPI Category	Total Value
0	Contacts KPI	861987672
1	Admissions KPI	3480883
2	Beds KPI	11697017
3	RTT KPI	795472
4	Crisis KPI	136566035

8. Source Mapping:

Source mapping was created to document how each KPI was extracted from the dataset. This mapping improves transparency and helps explain the filtering logic used during KPI extraction.

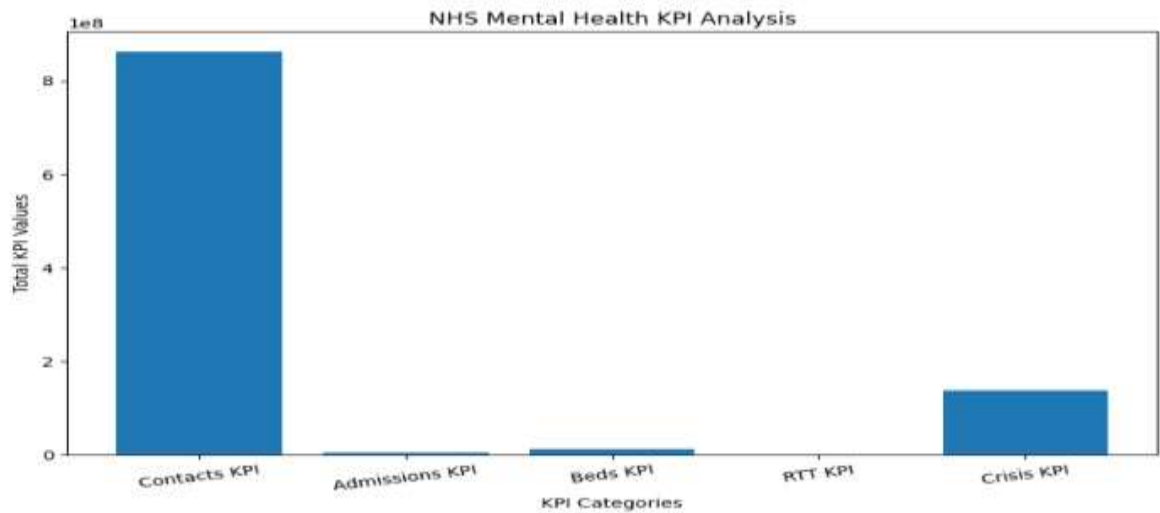
	KPI Category	Source Column	Keyword Filter
0	Contacts KPI	MEASURE_NAME	contact
1	Admissions KPI	MEASURE_NAME	inpatient admitted hospital
2	Beds KPI	MEASURE_NAME	bed ward
3	RTT KPI	MEASURE_NAME	waiting RTT
4	Crisis KPI	MEASURE_NAME	referral crisis

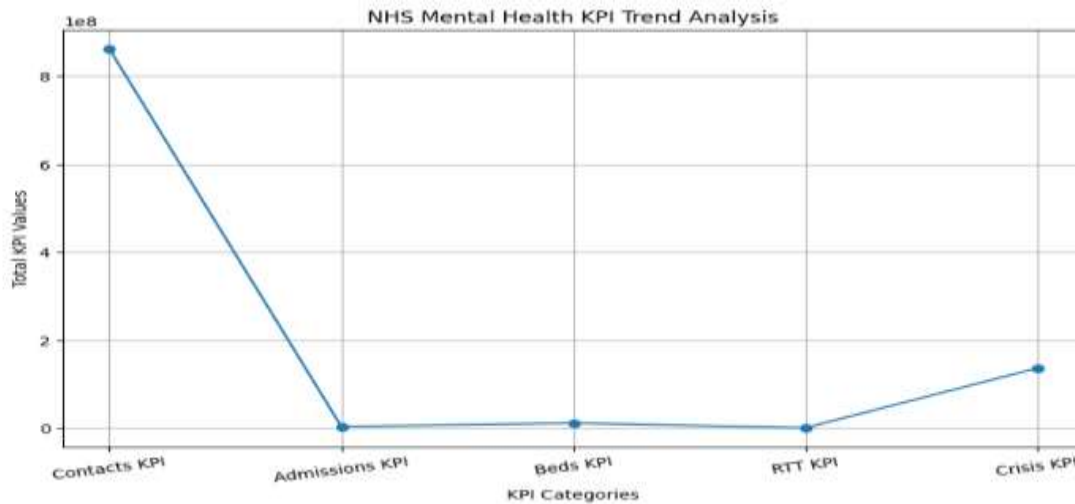
9. Visualizations:

Different charts were generated to visualize and compare NHS mental health KPI values.

Visualization Purpose:

- Bar charts were used to compare KPI values across different categories.
 - Horizontal bar charts improved readability for KPI comparison.
 - Line charts helped visualize KPI trends and differences more clearly.
- The visualizations made it easier to identify high-demand mental health service areas and compare NHS operational indicators.





10. Key Findings:

1. Contacts KPI recorded the highest value among all NHS mental health indicators, showing strong utilization of mental health support services across NHS systems.
2. Crisis-related measures also showed significantly high values, indicating increased demand for urgent mental health interventions and emergency care services.
3. Admissions KPI recorded comparatively lower values than contacts and crisis indicators, suggesting that inpatient care represents a smaller but critical portion of mental health services.
4. Bed-related metrics highlighted substantial inpatient mental health facility utilization and ward occupancy activity.
5. RTT (Referral to Treatment Time) measures indicated ongoing waiting-time pressure and referral delays within NHS mental health systems.

11. Conclusion:

The ANA-1 analysis successfully extracted and analysed five important NHS mental health KPIs using the official MHSDS dataset. The study provided insights into different operational areas of NHS mental health services, including patient contacts, admissions, crisis referrals, inpatient bed utilization, and treatment waiting times.

The analysis and visualizations helped identify major service trends and operational demand across NHS mental health systems. Overall, the task demonstrated how healthcare datasets can be processed and analysed using Python-based data analytics techniques to generate meaningful healthcare insights.